

Bill of Materials VBF-T10A-BOM-01

BOM

Component	Mass (g/cell)	Mass (kg/kWh)	Source note
Positive electrode active material (NMC811)	16.842	0.908	$\text{thickness} \times (1 - \text{porosity}) \times \text{density} \times \text{area}$ (layer_kg_m2)
Positive conductive additive (carbon black, 2 wt%)	0.337	0.018	literature default (no parameter)
Positive binder (PVDF, 2 wt%)	0.337	0.018	literature default (no parameter)
Negative electrode active material (graphite+SiOx)	10.874	0.586	$\text{thickness} \times (1 - \text{porosity}) \times \text{density} \times \text{area}$
Negative conductive additive (carbon black, 1 wt%)	0.109	0.006	literature default
Negative binder (SBR/CMC, 3 wt%)	0.326	0.018	literature default
Separator (polyolefin)	0.259	0.014	$\text{thickness} \times (1 - \text{porosity}) \times \text{density} \times \text{area}$
Electrolyte (1.2 g/cm3)	6.441	0.347	pore volume * 1.2 g/cm3 (literature density)
Positive current collector (Al)	4.437	0.239	$\text{thickness} \times \text{density} \times \text{area}$
Negative current collector (Cu)	11.042	0.595	$\text{thickness} \times \text{density} \times \text{area}$
Enclosure / tabs	Not modeled	Not modeled	Not modeled
Total (excl. enclosure)	51.004	2.75	cell energy 0.0186 kWh (energy_wh/1000)